

# Technical Guide (English)

Project: Download Bank of Taiwan Exchange Rate

Audience: Developers / DevOps / NetSuite engineers who need to change logic

Chinese version: TECHNICAL.zh-TW.md (./TECHNICAL.zh-TW.md)

User guide: USER\_GUIDE.en.md (./USER\_GUIDE.en.md)

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## 1. Goals and design principles

| Principle                           | Why                                                                  |
|-------------------------------------|----------------------------------------------------------------------|
| NetSuite must not call BOT directly | BOT uses bot protection; N/https often fails                         |
| Stable HTTPS via CDN                | Public GitHub repo + jsDelivr                                        |
| JSON first                          | NetSuite can JSON.parse without CSV parsing                          |
| Editable modules                    | Download, transform, schedule, and NetSuite write-back are separated |

Data flow:

```
rate.bot.com.tw CSV
  |
  ▼
scripts/download_bot_rates.py
- download with curl_cffi (browser TLS impersonation)
- CSV → JSON transform
- write data/
  |
  ▼
GitHub Actions (.github/workflows/update-rates.yml)
- weekday schedule / manual dispatch
- commit + push only when changed
- purge jsDelivr
  |
  ▼
cdn.jsdelivr.net/.../bot-xrt-latest.json
  |
  ▼
netsuite/DownloadBotRates_SS2.js
- N/https.get
- JSON.parse
- save File Cabinet (extendable to Currency Rate)
```

## 2. Repository layout

```
DownloadBankOfTaiwanExchangeRate/
├── .github/workflows/update-rates.yml # schedule + publish
├── data/
│   ├── bot-xrt-latest.csv           # latest raw CSV
│   ├── bot-xrt-latest.json          # latest JSON (CDN primary)
│   └── bot-xrt-latest.meta.txt       # checksum / fetch time
```

```
| └─ history/          # dated archives (last 90 days only)
|   └─ bot-xrt-YYYY-MM-DD.*
├─ docs/              # documentation
├─ netsuite/DownloadBotRates_SS2.js  # NetSuite Scheduled Script
├─ scripts/download_bot_rates.py     # download + transform core
├─ requirements.txt
└─ README.md
```

## 3. Local development

### 3.1 Environment

- Python 3.10+ (CI uses 3.12)
- Dependency: curl\_cffi (requirements.txt)

```
python3 -m pip install -r requirements.txt
python3 scripts/download_bot_rates.py
```

### 3.2 CLI flags

| Flag             | Purpose                                                |
|------------------|--------------------------------------------------------|
| -o / --output    | Path for bot-xrt-latest.csv; JSON is written beside it |
| --no-archive     | Skip dated archives under data/history/                |
| --retention-days | Keep only the latest N days in history (default 90)    |

### 3.3 Example success output

```
csv=.../data/bot-xrt-latest.csv
json=.../data/bot-xrt-latest.json
csv_bytes=3715
json_bytes=22669
rate_count=19
changed=true
archive_json=.../data/history/bot-xrt-2026-08-11.json
retention_days=90
history_migrated=0
history_pruned=0
```

History policy:

- Directory: data/history/
- Filenames: bot-xrt-YYYY-MM-DD.csv / .json
- Default retention: 90 days (older files are deleted automatically)
- Legacy dated files left under data/ are moved into history/ on each run

changed=false means content matched the previous files (meta still refreshes).

## 4. Core script: scripts/download\_bot\_rates.py

### 4.1 Functions (edit points)

| Function                  | Responsibility                     | Change when...                              |
|---------------------------|------------------------------------|---------------------------------------------|
| download_csv()            | Fetch BOT CSV                      | URL, headers, timeout, impersonate          |
| _to_rate()                | Parse numbers; map 0 → null        | Keep zeros, precision, filters              |
| _forward_block()          | Build forward tenors               | Add/remove days                             |
| csv_to_payload()          | CSV → JSON                         | Primary place to change JSON schema         |
| write_outputs()           | Write CSV/JSON/meta/history        | Filenames, archive policy, change detection |
| migrate_legacy_archives() | Move old dated files into history/ | Compatibility with previous layout          |
| prune_history()           | Delete files older than retention  | Retention policy changes                    |
| main()                    | CLI                                | Extra args / post-steps                     |

### 4.2 Key constants

```
BOT_CSV_URL = "https://rate.bot.com.tw/xrt/flcsv/0/day"
FORWARD_DAYS = (10, 30, 60, 90, 120, 150, 180)

BUY_CASH = 2
BUY_SPOT = 3
BUY_FWD_START = 4
SELL_CASH = 12
SELL_SPOT = 13
SELL_FWD_START = 14
```

### 4.3 BOT CSV column map (0-based)

Each data row is approximately:

| Index | Content                              |
|-------|--------------------------------------|
| 0     | Currency code (USD)                  |
| 1     | Label “本行買入” (bank buy)              |
| 2     | Cash buy                             |
| 3     | Spot buy                             |
| 4–10  | Forward buy 10/30/60/90/120/150/180  |
| 11    | Label “本行賣出” (bank sell)             |
| 12    | Cash sell                            |
| 13    | Spot sell                            |
| 14–20 | Forward sell 10/30/60/90/120/150/180 |

If BOT reshuffles columns, update constants + csv\_to\_payload() only. NetSuite can stay unchanged if the JSON contract remains stable.

### 4.4 Why curl\_cffi?

Plain requests / urllib TLS fingerprints are often challenged and return HTML.

curl\_cffi with impersonate="chrome" mimics a browser handshake.

If blocked again, try in order:

1. Switch impersonate (chrome / chrome131 / safari)
2. Adjust Referer / Accept-Language
3. Only then consider Playwright (heavier CI cost)

Keep these failure detectors:

- Content-Type contains text/html
- Body starts with <!DOCTYPE
- No comma in the first 200 bytes (not CSV-like)

## 4.5 JSON schema (contract)

```
{
  "source": "Bank of Taiwan",
  "sourceUrl": "https://rate.bot.com.tw/xrt/flcsv/0/day",
  "base": "TWD",
  "fetchAtUtc": "ISO-8601",
  "rateCount": 19,
  "rates": [ { "...": "CurrencyEntry" } ],
  "byCurrency": {
    "USD": { "...": "CurrencyEntry" }
  }
}
```

CurrencyEntry:

```
{
  "currency": "USD",
  "cash": { "buy": 31.87, "sell": 32.54 },
  "spot": { "buy": 32.195, "sell": 32.345 },
  "forward": {
    "buy": { "10": 32.214, "30": 32.164, "60": null, "90": null, "120": null, "150": null, "180": null },
    "sell": { "10": 32.318, "30": 32.272, "60": null, "90": null, "120": null, "150": null, "180": null }
  }
}
```

Contract rules:

- NetSuite expects both byCurrency and rates
- Missing quotes are null, not 0 (unless you update NetSuite checks together)
- Prefer additive fields (mid, localized names) for backward compatibility

## 4.6 Change detection

write\_outputs() compares byte equality of:

- bot-xrt-latest.csv
- bot-xrt-latest.json

Either difference ⇒ changed=true.

Note: fetchedAtUtc changes every run, so JSON often changes even when rates do not.  
To publish only when rates change, compare a payload without fetchedAtUtc (or compare canonical byCurrency).

Conceptual approach:

```
comparable = {k: v for k, v in payload.items() if k != "fetchedAtUtc"}  
new_bytes = json.dumps(comparable, ensure_ascii=False, sort_keys=True).encode()  
# compare against previous comparable; still write a fresh fetchedAtUtc when publishing
```

## 5. GitHub Actions: .github/workflows/update-rates.yml

### 5.1 Triggers

```
on:  
  workflow_dispatch:      # primary: external cron / manual  
  repository_dispatch:  
    types: [update-bot-rates]  
  schedule:               # backup only (UTC)  
    - cron: "30 2,7,10 * * 1-5" # Taiwan 10:30 / 15:30 / 18:30
```

Use an external cron as the primary trigger (GitHub's native schedule often skips).

Full setup: SCHEDULING.en.md (./SCHEDULING.en.md).

Local trigger:

```
./scripts/trigger_update.sh
```

### 5.2 Changing the schedule

| Need                 | Change                                            |
|----------------------|---------------------------------------------------|
| Change primary times | Edit cron-job.org (or your crontab)               |
| Change GitHub backup | Edit schedule.cron (UTC)                          |
| Manual only          | Disable external cron; optionally remove schedule |

### 5.3 Job steps and edit points

1. checkout
2. setup-python + pip install
3. run download\_bot\_rates.py
4. git diff on data/bot-xrt-latest.csv|json
5. commit / push only if changed
6. purge jsDelivr

Permissions:

```
permissions:  
  contents: write
```

Branch protection or custom deploy keys may require extra repo settings.

## 5.4 Manual CDN purge

```
curl -fsS "https://purge.jsdelivr.net/gh/ChesleyLo/DownloadBankOfTaiwanExchangeRate@main/data/bot-xrt-latest.json"
```

## 5.5 Repository must be public

jsDelivr cannot read private repos:

```
gh repo edit ChesleyLo/DownloadBankOfTaiwanExchangeRate \  
--visibility public \  
--accept-visibility-change-consequences
```

## 6. NetSuite: netsuite/DownloadBotRates\_SS2.js

### 6.1 Runtime

- Type: Scheduled Script 2.1
- Entry point: execute(context)
- Modules: N/https, N/file, N/log, N/runtime

### 6.2 Constants (commonly edited)

| Constant           | Meaning                         |
|--------------------|---------------------------------|
| CDN_JSON_URL       | Default CDN endpoint            |
| FOLDER_ID          | File Cabinet folder internal ID |
| FILE_NAME          | Latest filename                 |
| DEFAULT_RATE_FIELD | e.g. spot.sell                  |

### 6.3 Functions (edit points)

| Function           | Responsibility                  | Change when...                 |
|--------------------|---------------------------------|--------------------------------|
| getConfiguredUrl() | Script param or default URL     | Multi-environment URLs         |
| getRateFieldPath() | cash/spot selection             | Accounting policy change       |
| parsePayload()     | Validate JSON contract          | Schema changes                 |
| readRate()         | Resolve dotted paths            | Support forward.sell.30 etc.   |
| saveJson()         | Write File Cabinet              | Naming / overwrite policy      |
| maybeApplyRates()  | Rate write-back extension point | Currency Rate / custom records |
| execute()          | Main flow                       | Alerts, retries, fallbacks     |

### 6.4 Recommended script parameters

| ID | Type | Purpose |
|----|------|---------|
|----|------|---------|

|                             |          |                         |
|-----------------------------|----------|-------------------------|
| custscript_bot_cdn_json_url | Text     | CDN URL                 |
| custscript_bot_rate_field   | Text     | spot.sell / cash.buy... |
| custscript_bot_apply_rates  | Checkbox | Enable write-back       |

## 6.5 Enabling real Currency Rate writes

maybeApplyRates() currently only log.debugs. To persist rates:

1. Add "N/record" (and N/search / N/format if needed)
2. Map fields for your account's Currency Rate record
3. Handle multi-currency base, effective date, and inverse rates (TWD→USD vs USD→TWD)

Conceptual sketch (must be adapted to your account; do not copy blindly):

```
define(["N/https", "N/file", "N/log", "N/runtime", "N/record"], (
  https, file, log, runtime, record
) => {
  // ...
  function upsertCurrencyRate(fromCode, toCode, exchangeRate, effectiveDate) {
    const rec = record.create({ type: record.Type.CURRENCY_RATE, isDynamic: true });
    // rec.setValue({ fieldId: "...", value: ... }); // account-specific
    return rec.save();
  }
});
```

readRate() already supports dotted paths such as forward.sell.30.

## 6.6 Restrict currencies

Inside maybeApplyRates():

```
const ALLOWED = ["USD", "EUR", "JPY", "CNY"];
currencies.filter((c) => ALLOWED.indexOf(c) !== -1).forEach(/* ... */);
```

## 6.7 Error-handling ideas

Already present:

- Non-2xx HTTP → throw
- Invalid JSON → throw
- Missing byCurrency/rates → throw

Optional enhancements:

- Retry 2–3 times for transient CDN failures
- Email / custom record on failure
- Fallback to GitHub Raw URL

## 7. Cookbook

## 7.1 Add mid = (buy + sell) / 2

After building entry in csv\_to\_payload():

```
def _mid(side: dict) -> float | None:
    b, s = side.get("buy"), side.get("sell")
    if b is None or s is None:
        return None
    return round((b + s) / 2, 6)

entry["spot"]["mid"] = _mid(entry["spot"])
entry["cash"]["mid"] = _mid(entry["cash"])
```

NetSuite can then read spot.mid.

## 7.2 Historical date CSV

BOT may support:

```
https://rate.bot.com.tw/xrt/flcsv/0/YYYY-MM-DD
```

Parameterize:

```
parser.add_argument("--date", help="YYYY-MM-DD; default=day")
# url = f"https://rate.bot.com.tw/xrt/flcsv/0/{args.date or 'day'}"
```

## 7.3 Publish to another CDN / object store

Replace the final Actions step with S3/R2/Azure upload and update the NetSuite URL.  
Keep the JSON contract stable so SuiteScript changes stay minimal.

## 7.4 Rename published files

Update all of:

1. write\_outputs() filenames
2. Actions purge paths
3. NetSuite CDN\_JSON\_URL / FILE\_NAME
4. URLs in user docs

---

## 8. Test checklist

Download / transform

- [ ] python3 scripts/download\_bot\_rates.py succeeds
- [ ] JSON loads with json.load
- [ ] byCurrency.USD.spot.sell is a number
- [ ] Missing quotes are null
- [ ] Blocked/offline run exits non-zero



## Actions

- [ ] Manual workflow run succeeds
- [ ] Commit updates data/bot-xrt-latest.json
- [ ] CDN URL returns HTTP 200
- [ ] After purge, CDN serves the new content

## NetSuite

- [ ] Execute Now succeeds
  - [ ] File Cabinet has latest + dated files
  - [ ] Execution Log shows USD spot sell
  - [ ] (If enabled) Currency Rate / custom records are correct
- 

## 9. Troubleshooting

| Symptom                       | Likely cause              | Action                               |
|-------------------------------|---------------------------|--------------------------------------|
| HTML Challenge downloaded     | WAF                       | Verify curl_cffi, switch impersonate |
| CDN 404 file not found        | Private repo / wrong path | Make public; verify path             |
| Stale CDN content             | Cache                     | Purge; temporarily use Raw URL       |
| Actions cannot push           | Token/permissions         | contents: write, branch rules        |
| NetSuite connectivity failure | Allowlist                 | Permit cdn.jsdelivr.net              |
| JSON parse error              | HTML/error page returned  | Check URL and HTTP status            |

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## 10. Security and compliance

- Never commit secrets to a public repo
  - Cite Bank of Taiwan as the source
  - CDN is a mirror and may lag; define reconciliation for finance-critical use
  - Rate write-back is sensitive — require code review
- 

## 11. Maintenance tips

1. Evolve the JSON contract additively (add fields before removing)
  2. Update bilingual docs + README for material changes
  3. If BOT changes CSV layout, fix Python first and confirm NetSuite still works
  4. Quarterly spot-check CDN values against the official BOT page
- 

## 12. Document index

| Document                                    | Description                    |
|---------------------------------------------|--------------------------------|
| TECHNICAL.zh-TW.md (./TECHNICAL.zh-TW.md)   | Technical guide (Chinese)      |
| TECHNICAL.en.md (./TECHNICAL.en.md)         | This technical guide (English) |
| USER_GUIDE.zh-TW.md (./USER_GUIDE.zh-TW.md) | User guide (Chinese)           |
| USER_GUIDE.en.md (./USER_GUIDE.en.md)       | User guide (English)           |